Healthcare Appointment Booking Application

Domain: Manage Appointment

Use Case Name: Healthcare Appointment

Use Case Description:

This use case outlines the process of **registering, logging in, searching, booking, viewing, updating, and managing appointments** through a healthcare appointment booking system with **three modules**: **Patient**, **Hospital**, and **Doctor**.

- **Patients** can search and book appointments.
- **Hospitals** manage doctor availability and appointments.
- **Doctors** view their schedules and access patient details.

Use Case Goal:

To enable seamless communication between patients, hospitals, and doctors, offering an efficient and transparent way to manage appointments and healthcare services.

Actors:

- **Patient**
- **Hospital/Administrator**
- **Doctor**

Pre-Conditions:

- Patient, hospital, and doctor accounts are registered.
- The patient is logged into the system.
- The hospital has registered available doctors and time slots.
- Doctor availability is updated in the system.

Post-Conditions:

- Appointments are successfully booked, modified, viewed, or cancelled.
- System notifications are triggered to all stakeholders.
- Doctors can view their appointment schedule and patient details.

Main Flow:

1. Patient Flow:

1. Patient logs in to the application.
2. Searches for hospitals by location or specialization.
3. Filters doctors based on availability and specialization.
4. Selects a doctor and time slot.
5. Reviews and confirms the appointment.
6. Receives booking confirmation and notification.

2. Hospital Flow:

1. Hospital logs in to the system.
2. Adds new doctors to the system (name, specialization,slots).
3. Sets doctor availability via time slot management.
4. Views appointment list and manages booking requests.

3. Doctor Flow:

1. Doctor registers and logs in.
2. Views their personal dashboard showing upcoming appointments.
3. Accesses each patient's profile and medical history (if available).
4. Marks appointments as completed or updates status.

Alternative Flows:

- **No Available Slots** → Patient added to waitlist or notified.
- **Rescheduling** → Patient or doctor reschedules appointment.
- **Cancellation** → Appointment cancelled, slot reopened.
- **Doctor Not Available** → System notifies hospital and patient.

Exception Handling:

1.Network Lost -Shows retry option

2.Doctor Unavailable -Suggests alternatives

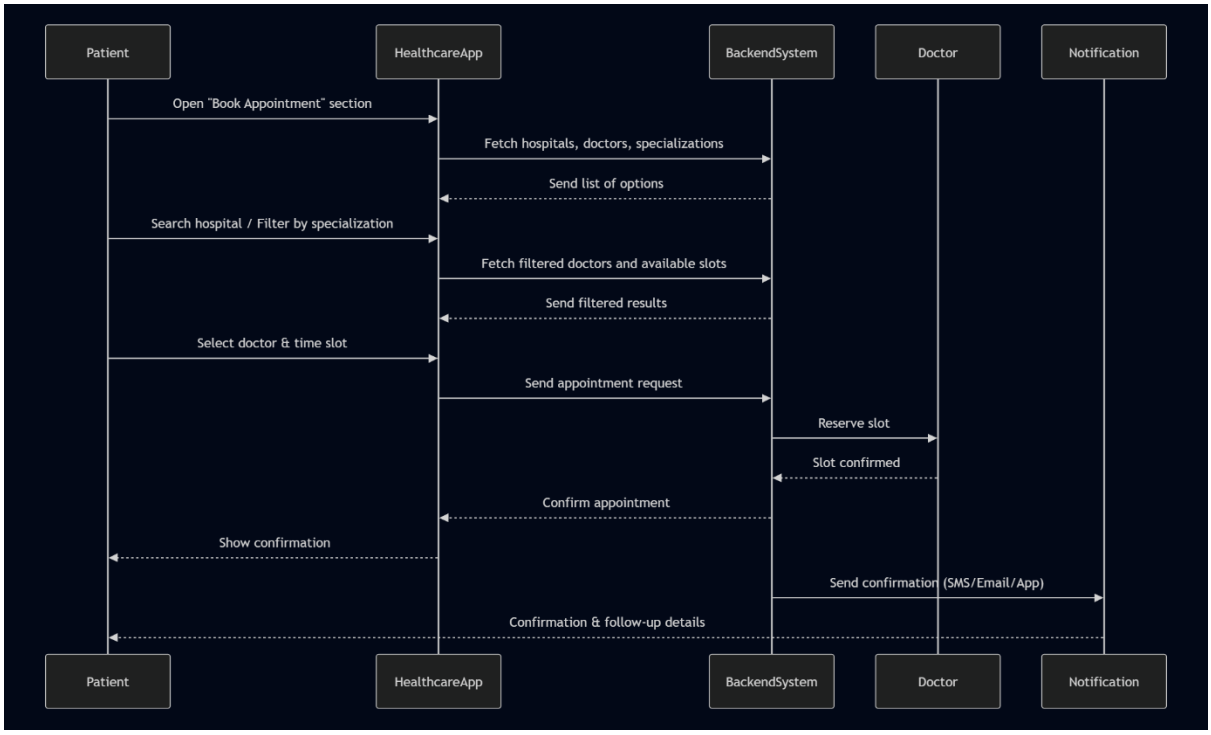
3.Slot Conflict -Prevents double-booking

Test Cases:

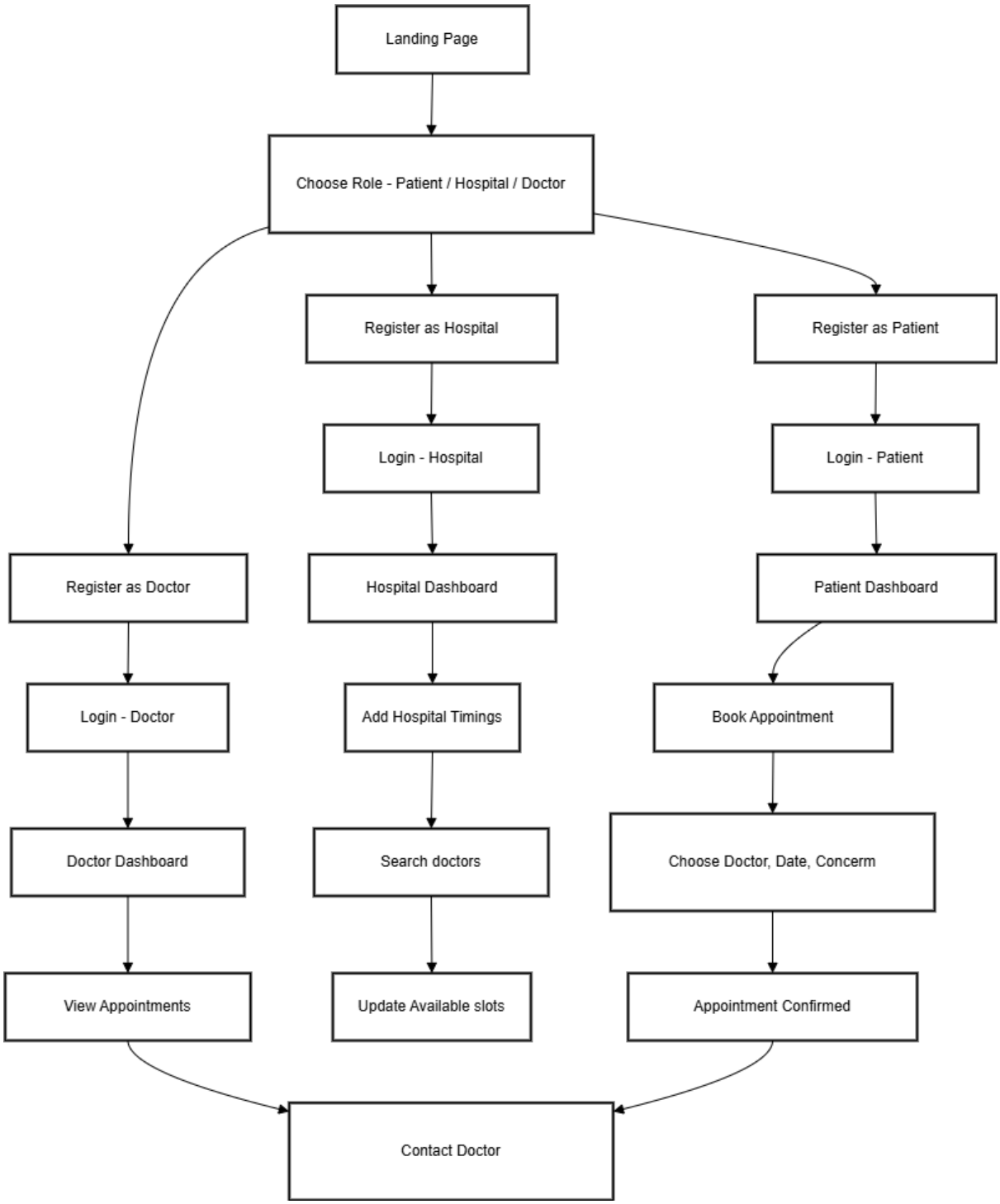
Test Case ID	Description	Input Data	Expected Outcome
TC001	Register a new patient	Name, email, phone, password	Account created
TC002	Existing email registration	Registered email	Error shown
TC003	Patient login	Email & password	Login success
TC004	Incorrect login	Wrong credentials	Error message
TC005	Booking without login	None	Prompt login
TC006	Search hospitals	Location input	Hospital list
TC007	Filter doctors	Specialization	Filtered list
TC008	View slots	Doctor selected	Available slots shown
TC009	Book appointment	Doctor, slot	Booking success
TC010	Cancel appointment	Appointment ID	Cancellation success
TC011	Book already booked slot	is_booked = true	Show “Unavailable”
TC012	Track appointment	Appointment ID	Status shown
TC013	Reschedule appointment	Appointment ID, new slot	Rescheduled
TC014	Add availability	Doctor ID, time slot	Slot added
TC015	View past appointments	Patient logged in	History shown
TC016	Book telehealth appointment	Virtual slot	Secure link
TC017	Logout	Logout clicked	Session ended
TC018	Register doctor	Name, email, specialization, password	Doctor account created
TC019	Doctor login	Email, password	Dashboard shown
TC020	Doctor views schedule	Logged-in doctor	Appointments list
TC021	Doctor views patient details	Appointment ID	Patient info shown

Test Case ID	Description	Input Data	Expected Outcome
TC022	Doctor marks appointment complete	Appointment ID	Status updated
TC023	Doctor not available	Unavailability submitted	Patient notified

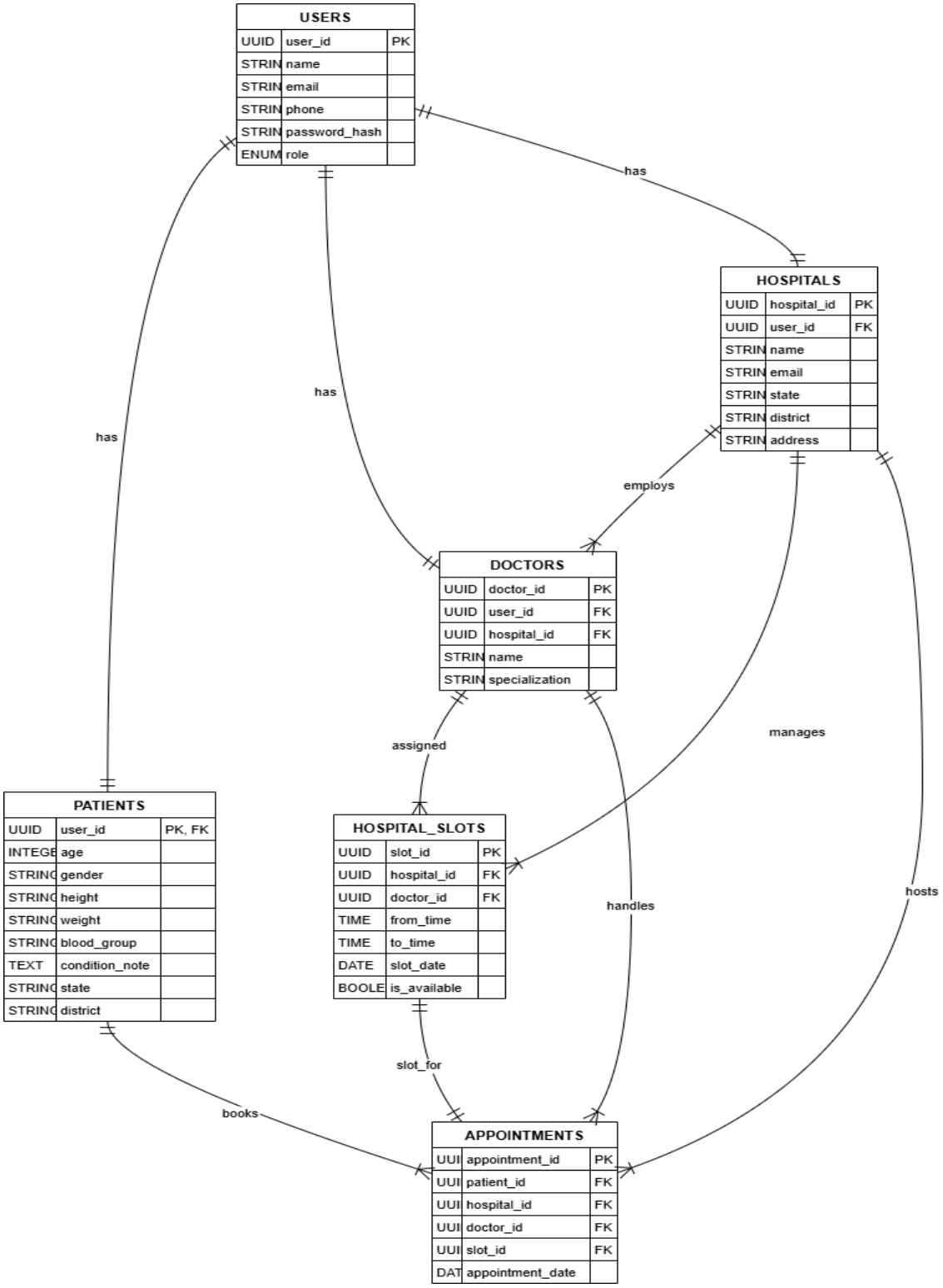
Sequence Diagram



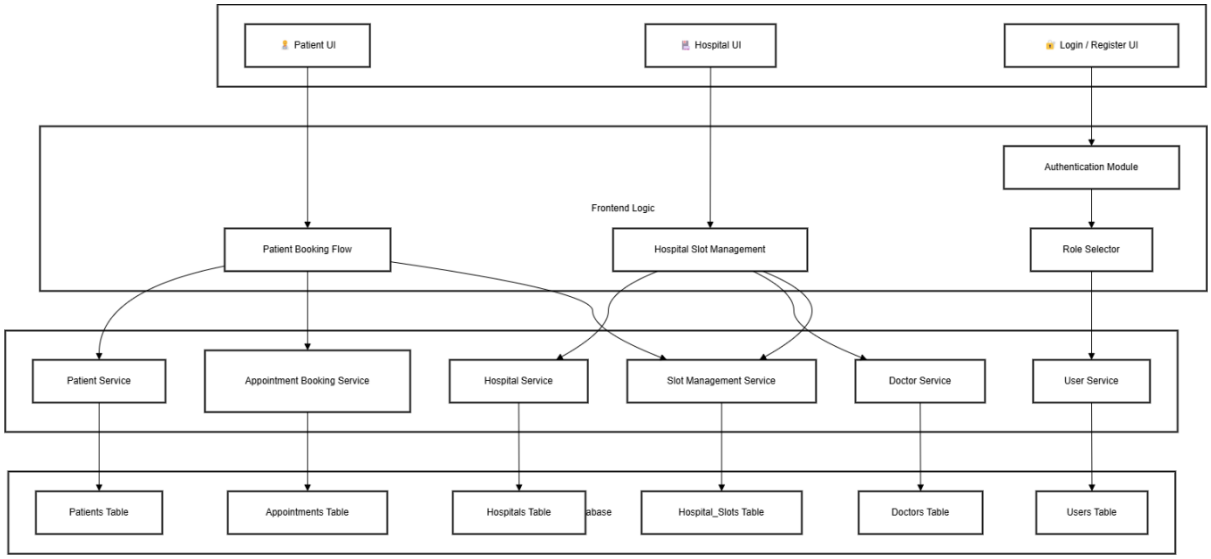
Workflow Design



Database Design Diagram :



Functional Architecture :



Expected Outcome:

- Patients can easily manage appointments.
- Hospitals streamline doctor and slot management.
- Doctors can handle appointments and track patient data.
- Secure, user-friendly interactions with robust notification system.